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== Vitis HLS Report for 'matVecMul'

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* Date: Mon May 4 03:03:57 2026

* Version: 2021.1 (Build 3247384 on Thu Jun 10 19:36:07 MDT 2021)
* Project: matVecMul_stream_prj
* Solution: solution2 (Vivado IP Flow Target)
* Product family: zynq
* Target device: xc7z020-clg400-1

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== Performance Estimates

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+ Timing:
  * Summary:

Clock	Target	Estimated	Uncertainty
ap_clk	10.00 ns	5.831 ns	2.70 ns

+ Latency:
  * Summary:

Latency (cycles)		Latency (absolute)		Interval		Pipeline
min	max	min	max	min	max	Type
4870	4870	48.700 us	48.700 us	4871	4871	no

  + Detail:
    * Instance:

		Latency (cycles)		Latency (absolute)		Interval		Pipeline
Instance	Module	min	max	min	max	min	max	Type
grp_matVecMul_Pipeline_sample_loop_fu_190	matVecMul_Pipeline_sample_loop	66	66	0.660 us	0.660 us	66	66	no
grp_matVecMul_Pipeline_matmul_inner_loop_fu_211	matVecMul_Pipeline_matmul_inner_loop	70	70	0.700 us	0.700 us	70	70	no

    * Loop:

	Latency (cycles)		Iteration	Initiation Interval		Trip	
Loop Name	min	max	Latency	achieved	target	Count	Pipelined
- matmul_outer_loop	4800	4800	75	-	-	64	no

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== Utilization Estimates

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* Summary:

Name	BRAM_18K	DSP	FF	LUT	URAM
DSP	-	-	-	-	-
Expression	-	-	0	238	-
FIFO	-	-	-	-	-
Instance	4	1	354	609	-
Memory	1	-	0	0	0
Multiplexer	-	-	-	117	-

Register	-	-	152	-	-
Total	5	1	506	964	0
Available	280	220	106400	53200	0
Utilization (%)	1	~0	~0	1	0

+ Detail:

* Instance:

Instance	Module	BRAM_18K	DSP	FF	LUT	URAM
control_s_axi_U	control_s_axi	4	0	192	238	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_211	matVecMul_Pipeline_matmul_inner_loop	0	1	153	299	0
grp_matVecMul_Pipeline_sample_loop_fu_190	matVecMul_Pipeline_sample_loop	0	0	9	72	0
Total		4	1	354	609	0

* DSP:

N/A

* Memory:

Memory	Module	BRAM_18K	FF	LUT	URAM	Words	Bits	Banks	W*Bits*Banks
x_localbuff_V_U	x_localbuff_V	1	0	0	0	64	16	1	1024
Total		1	0	0	0	64	16	1	1024

* FIFO:

N/A

* Expression:

Variable Name	Operation	DSP	FF	LUT	Bitwidth P0	Bitwidth P1
add_ln34_fu_240_p2	+	0	0	39	32	1
p_Val2_s_fu_309_p2	+	0	0	23	16	16
sub_fu_227_p2	+	0	0	39	32	2
and_ln789_fu_395_p2	and	0	0	2	1	1
and_ln790_fu_409_p2	and	0	0	2	1	1
and_ln795_fu_439_p2	and	0	0	2	1	1
carry_1_fu_329_p2	and	0	0	2	1	1
overflow_fu_433_p2	and	0	0	2	1	1
underflow_fu_457_p2	and	0	0	2	1	1
Range1_all_ones_fu_361_p2	icmp	0	0	12	14	2
Range1_all_zeros_fu_367_p2	icmp	0	0	12	14	1
Range2_all_ones_fu_345_p2	icmp	0	0	12	13	2
icmp_ln34_fu_235_p2	icmp	0	0	18	32	32
out_pkt_last_V_fu_250_p2	icmp	0	0	18	32	32
or_ln384_fu_471_p2	or	0	0	2	1	1
or_ln794_fu_421_p2	or	0	0	2	1	1
or_ln795_fu_445_p2	or	0	0	2	1	1
deleted_ones_fu_401_p3	select	0	0	2	1	1
deleted_zeros_fu_373_p3	select	0	0	2	1	1
select_ln384_fu_463_p3	select	0	0	17	1	15
val_V_fu_477_p3	select	0	0	16	1	16
xor_ln416_fu_323_p2	xor	0	0	2	1	2
xor_ln789_fu_389_p2	xor	0	0	2	1	2
xor_ln794_1_fu_427_p2	xor	0	0	2	1	2
xor_ln794_fu_415_p2	xor	0	0	2	1	2
xor_ln795_fu_451_p2	xor	0	0	2	1	2

Total			0	0	238	203	140

* Multiplexer:

Name	LUT	Input Size	Bits	Total Bits
ap_NS_fsm	53	10	1	10
i_1_fu_140	9	2	32	64
in_stream_TREADY_int_regslice	9	2	1	2
out_stream_TDATA_blk_n	9	2	1	2
x_localbuff_V_address0	14	3	6	18
x_localbuff_V_ce0	14	3	1	3
x_localbuff_V_we0	9	2	1	2
Total	117	24	43	101

* Register:

Name	FF	LUT	Bits	Const Bits
ap_CS_fsm	9	0	9	0
col_size_read_reg_497	32	0	32	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_211_ap_start_reg	1	0	1	0
grp_matVecMul_Pipeline_sample_loop_fu_190_ap_start_reg	1	0	1	0
i_1_fu_140	32	0	32	0
out_pkt_last_V_reg_528	1	0	1	0
row_size_read_reg_503	32	0	32	0
shl_ln_reg_533	6	0	13	7
sub_reg_515	32	0	32	0
trunc_ln1169_reg_523	6	0	6	0
Total	152	0	159	7

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== Interface
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* Summary:

RTL Ports	Dir	Bits	Protocol	Source Object	C Type
s_axi_control_AWVALID	in	1	s_axi	control	array
s_axi_control_AWREADY	out	1	s_axi	control	array
s_axi_control_AWADDR	in	14	s_axi	control	array
s_axi_control_WVALID	in	1	s_axi	control	array
s_axi_control_WREADY	out	1	s_axi	control	array
s_axi_control_WDATA	in	32	s_axi	control	array
s_axi_control_WSTRB	in	4	s_axi	control	array
s_axi_control_ARVALID	in	1	s_axi	control	array
s_axi_control_ARREADY	out	1	s_axi	control	array
s_axi_control_ARADDR	in	14	s_axi	control	array
s_axi_control_RVALID	out	1	s_axi	control	array
s_axi_control_RREADY	in	1	s_axi	control	array
s_axi_control_RDATA	out	32	s_axi	control	array
s_axi_control_RRESP	out	2	s_axi	control	array
s_axi_control_BVALID	out	1	s_axi	control	array
s_axi_control_BREADY	in	1	s_axi	control	array
s_axi_control_BRESP	out	2	s_axi	control	array
ap_clk	in	1	ap_ctrl_hs	matVecMul	return value
ap_rst_n	in	1	ap_ctrl_hs	matVecMul	return value
interrupt	out	1	ap_ctrl_hs	matVecMul	return value
in_stream_TDATA	in	32	axis	in_stream_V_data_V	pointer
in_stream_TVALID	in	1	axis	in_stream_V_dest_V	pointer

in_stream_TREADY	out	1	axis	in_stream_V_dest_V	pointer
in_stream_TDEST	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TKEEP	in	4	axis	in_stream_V_keep_V	pointer
in_stream_TSTRB	in	4	axis	in_stream_V_strb_V	pointer
in_stream_TUSER	in	1	axis	in_stream_V_user_V	pointer
in_stream_TLAST	in	1	axis	in_stream_V_last_V	pointer
in_stream_TID	in	1	axis	in_stream_V_id_V	pointer
out_stream_TDATA	out	32	axis	out_stream_V_data_V	pointer
out_stream_TVALID	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TREADY	in	1	axis	out_stream_V_dest_V	pointer
out_stream_TDEST	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TKEEP	out	4	axis	out_stream_V_keep_V	pointer
out_stream_TSTRB	out	4	axis	out_stream_V_strb_V	pointer
out_stream_TUSER	out	1	axis	out_stream_V_user_V	pointer
out_stream_TLAST	out	1	axis	out_stream_V_last_V	pointer
out_stream_TID	out	1	axis	out_stream_V_id_V	pointer